ALY 3510 AI Foundations: An Interdisciplinary Approach

Week 8:

Feedback to week 7

Session/Section: Fall 2026
Instructor: Vladimir Shapiro

Northeastern University

Feedback - Week 7
Midterm

Big discrepancies:

- Some 100-point grades, but some much lower ones

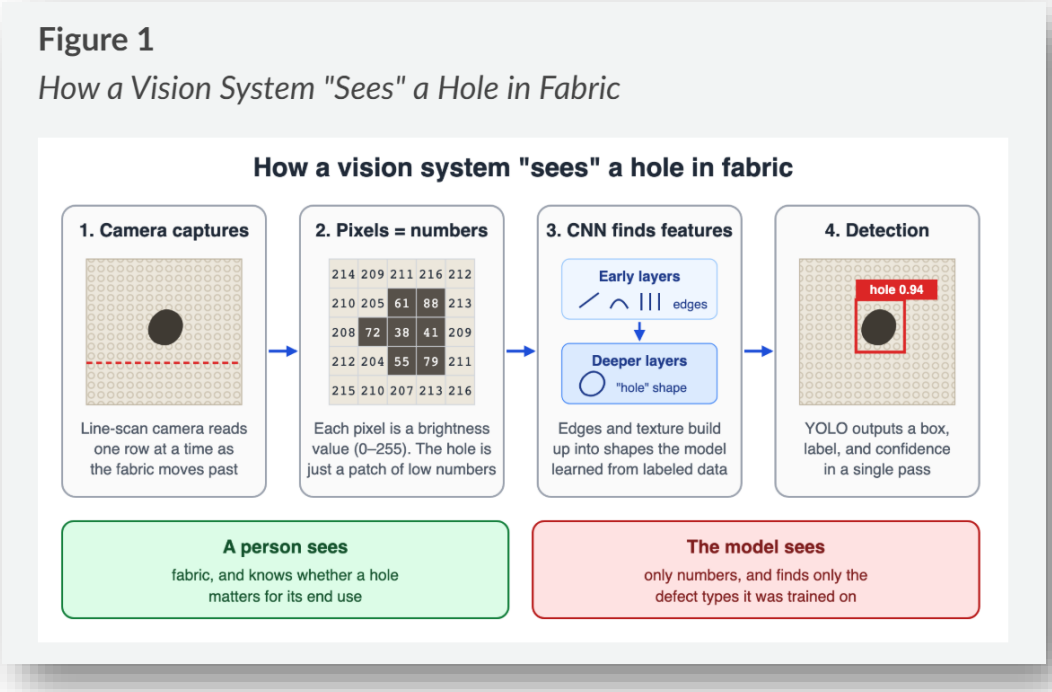
Feedback - Week 7

Reflection board



Going well:

- Good understanding of the subject and challenges
- Helpful visuals: Nathaniel, Jennifer



- Mohammad: "I learned this week is that recognizing something isn't necessarily the same as understanding it"

Feedback - Week 7

Reflection board

To improve:

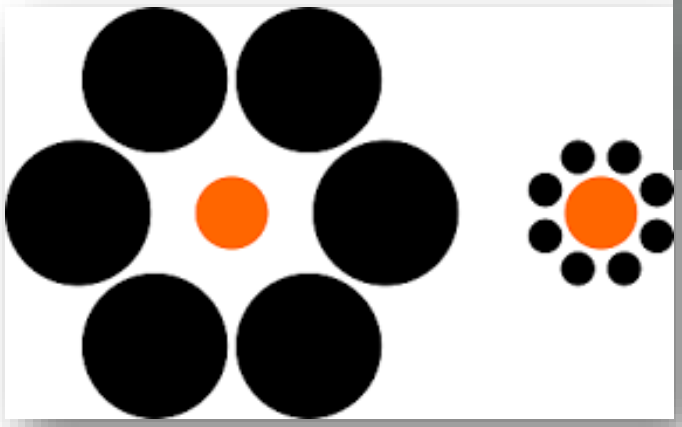
- Reflecting on computer vision without showing visuals is problematic
- "A doctor naturally understands that a blurry spot might just be an imaging artifact from a coughing patient" – is it a sign of human vision superiority or doctor training?
- "Human vision works differently; we are able to recognize faces and aren't fooled by layers of noise or pixels." - I'm not sure about the elderly or toddlers. Even adults struggle to recognize faces across races.

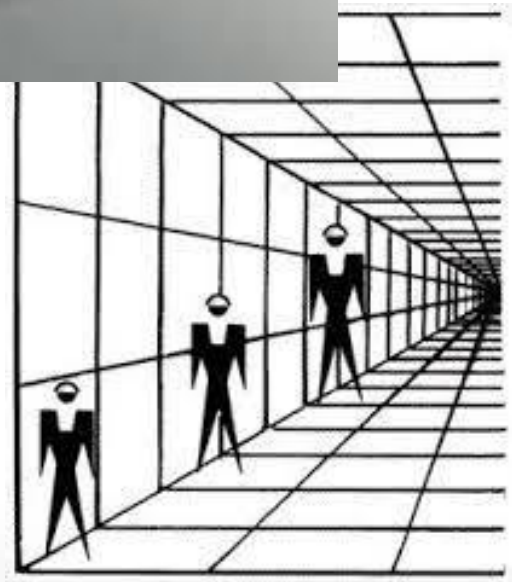
Miscellaneous

Human vision imperfections

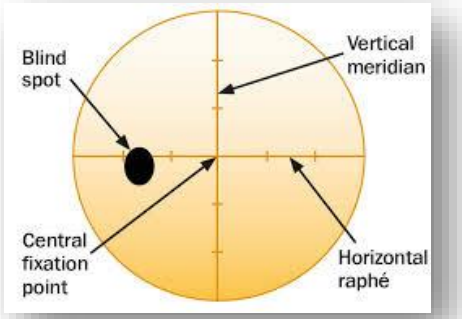
- Optical illusions:







- Vision in the dark is severely restricted
- Can't detect too slow or too fast changes
- Spatial resolution is limited
- Color discrimination is restricted
- Blind spot
- How many wear glasses?



Topics Covered

Course Tracker

Week #	Topic
1	Overview of AI, history, definitions, and key concepts
2	Knowledge Representation and Reasoning
3	Introduction to Machine Learning
4	Neural Networks & Deep Learning
5	Distributed AI & AI Platforms
6	Natural Language Processing (NLP)
7	Computer Vision
8	**Robotics & Embodied AI**
9	AI & Creativity
10	Applications of AI – Healthcare, Finance, Education, Retail, Marketing, Entertainment
11	Applications of AI – Biomedical, Agriculture, Transportation, Mining, Manufacturing, Energy
12	AI and Society, Bias & Fairness in AI
13	The Future of AI
14	No Lecture - Final Project Report & Presentation are Due